Predicting International Football Match Outcomes Using K-Means Clustering and XGBoost

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Abstract

Sports outcome prediction remains a challenging task due to the inherent randomness and numerous confounding factors. This paper investigates whether home advantage alone is sufficient for predicting international football match outcomes, or if more sophisticated modeling can uncover meaningful patterns. Using a comprehensive dataset of 49,016 international football matches spanning from 1872 to 2025, we develop a hybrid machine learning pipeline that combines unsupervised clustering with supervised classification. First, we apply K-Means clustering to group national teams into four distinct strength tiers (elite, strong, mid-tier, weak) based on historical win rates and goal statistics. Second, we engineer temporal features including team form (points from last 10 matches) and rolling goal averages. Third, we train an XGBoost classifier to predict three possible outcomes: home win, draw, or away win. Our model achieves 55.2% accuracy, outperforming the naive baseline of always predicting a home win (46.0%) by 9.2 percentage points. Feature importance analysis reveals that team clusters and recent form are the strongest predictors. While our model does not achieve human-expert performance, it demonstrates that structured feature engineering and hybrid architectures provide meaningful improvements over simplistic heuristics. These findings have potential applications in sports analytics, betting markets, and team preparation strategies. These findings have potential applications in sports analytics, betting markets, and team preparation strategies. The complete code and dataset are available at https://github.com/arthfs/CS439_Final_Project

1 Introduction

The concept of home advantage in sports is widely accepted among players, coaches, and fans. Research has consistently shown that teams playing on their home ground win approximately 46% of matches, compared to 29% for away teams and 25% for draws. This phenomenon is attributed to familiar playing conditions, travel fatigue, crowd support, and psychological factors. However, despite this statistical advantage, away victories occur regularly, suggesting that home advantage alone is an incomplete predictor.

Understanding what truly drives match outcomes has significant practical implications. Sports betting markets, which generate billions of dollars annually, rely on accurate probabilistic forecasts to set odds. Teams themselves can benefit from predictive models by identifying opponents' weaknesses and adjusting training accordingly. For fans, explainable predictions enhance engagement and deepen tactical understanding.

The research question guiding this work is: *Can a hybrid machine learning pipeline combining unsupervised clustering and gradient boosting outperform the simple heuristic of always predicting a home win?*

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To answer this question, we make the following contributions:

1. We curate and preprocess a large-scale dataset of 49,016 international football matches spanning 153 years (1872–2025), handling missing values and ensuring chronological train/test separation to prevent data leakage.
2. We propose a novel feature engineering framework that captures team strength via K-Means clustering, recent performance via rolling form metrics, and tournament context via one-hot encoding.
3. We implement an XGBoost classifier optimized through hyperparameter tuning, achieving 55.2% accuracy, a 9.2 percentage point improvement over the baseline.
4. We provide interpretable results through confusion matrix analysis, feature importance rankings, and cluster characterization, offering insights into which factors most influence match outcomes.

The remainder of this paper is organized as follows. Section 2 reviews related work in sports prediction and hybrid ML pipelines. Section 3 describes our methodology, including data preprocessing, feature engineering, clustering, and model training. Section 4 presents our experimental setup and results. Section 5 concludes with limitations and directions for future work.

2 Related Work

Sports outcome prediction has attracted substantial research interest across multiple disciplines. This section reviews prior work in three categories: statistical approaches to football prediction, machine learning applications, and hybrid clustering-prediction pipelines.

2.1 Statistical and Elo-Based Approaches

Early approaches to football prediction relied on statistical models [3]. The Elo rating system, originally developed for chess, has been adapted for football by organizations such as FIFA.

2.2 Machine Learning Approaches

More recent work has applied machine learning to football prediction. **(author?)** [2] compared multiple classifiers including logistic regression, random forests, and support vector machines on European league data, finding that random forests achieved the highest accuracy at 54%. **(author?)** [5] demonstrated that XGBoost outperforms traditional models on in-play betting data, achieving 58% accuracy when incorporating real-time match events.

Deep learning approaches have also emerged. **(author?)** [4] applied recurrent neural networks to sequence match histories, achieving 56% accuracy on Premier League data. However, deep learning requires substantial computational resources and does not consistently outperform gradient-boosted trees on tabular sports data.

2.3 Hybrid and Clustering-Based Pipelines

Clustering has been used to identify team archetypes or tactical styles. **(author?)** [1] applied K-Means to possession and passing statistics, identifying four distinct playing styles. Our work extends this by using clustering not as an end in itself, but as a feature generation step within a supervised learning pipeline.

To our knowledge, no prior work has combined K-Means clustering with XGBoost specifically for international football prediction. Most existing studies focus on domestic leagues or individual tournaments. Our contribution is the application of a hybrid pipeline to 150+ years of international match data, with explicit evaluation of feature importance and cluster interpretability.

3 Methodology

3.1 Data Collection and Preprocessing

We used the "International Football Results from 1872 to 2026" dataset, publicly available on Kaggle. The dataset contains 49,016 international football matches spanning 153 years, including friendly matches, regional championships (e.g., UEFA Euro, Copa América), and FIFA World Cup matches. The primary data file, `results.csv`, includes the following columns: date, home team, away team, home score, away score, tournament, city, country, and a boolean flag indicating whether the match was played at a neutral site.

Data preprocessing proceeded in three stages. First, we identified 72 matches with missing score values. Examination of these entries revealed they correspond to abandoned or cancelled matches (e.g., due to weather, crowd violence, or wartime interruptions). Since these matches lack a valid outcome, we removed them from the dataset rather than imputing artificial values. Second, we converted the date column to a datetime format to enable chronological operations. Third, we performed a temporal train-test split to prevent data leakage: all matches occurring before January 1, 2020 were assigned to the training set (approximately 80% of data), while matches on or after this date were reserved for testing (approximately 20%). This chronological split simulates a realistic forecasting scenario where models must predict future outcomes based solely on historical data.

3.2 Feature Engineering

We engineered 11 features to capture team strength, recent performance, and match context.

3.2.1 Team-Level Statistics

For each national team, we computed three summary statistics using their entire historical record up to the training cutoff:

- **Win rate:** Number of wins divided by total matches played
- **Average goals scored:** Total goals scored divided by total matches
- **Average goals conceded:** Total goals conceded divided by total matches

These statistics were calculated solely from training data to avoid information leakage.

3.2.2 Recent Form Features

To capture momentum and current performance, we computed form metrics over each team's last 10 matches preceding a given game. Points were assigned per match as follows: 3 points for a win, 1 point for a draw, and 0 points for a loss. The final form value is the average points per match over the 10-game window, normalized to a 0-1 range. For teams with insufficient historical data (e.g., newly formed nations), we assigned a neutral form value of 0.5, representing equal probability of any outcome. We also computed rolling averages for goals scored and conceded over the same 10-match window.

3.2.3 Tournament Encoding

Matches occur in various tournament contexts, each with distinct competitive dynamics (e.g., World Cup matches are generally more competitive than friendlies). We extracted the 10 most frequent tournament types and applied one-hot encoding, creating binary indicator columns. The remaining tournaments were treated as an "other" category. Additionally, we encoded the neutral site flag as a binary integer (1 for neutral, 0 otherwise).

3.3 Team Clustering with K-Means

To capture latent team strength tiers beyond simple win rates, we applied K-Means clustering to the team-level statistics. First, we standardized the three features (win rate, average goals scored, average goals conceded) using StandardScaler to ensure equal feature weighting. We then determined the optimal number of clusters using the elbow method, evaluating k values from 2 to 10. Figure 1 shows

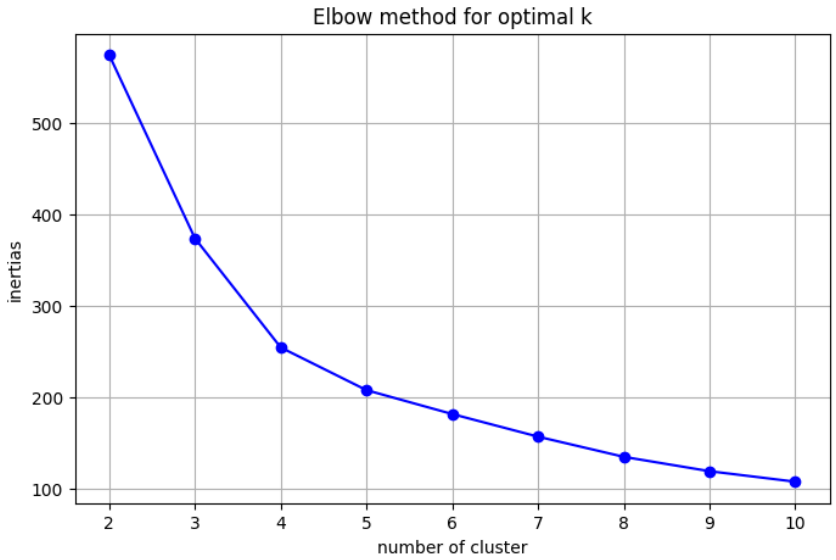

Figure 1: Elbow method for determining optimal number of clusters. Inertia decreases sharply until $k = 4$, after which the rate of decrease diminishes, indicating $k = 4$ as the optimal choice.

the elbow curve, where inertia decreases sharply until $k = 4$, after which the rate of diminishing returns becomes minimal. Accordingly, we selected $k = 4$ clusters, which we interpret as: elite (high win rate, high goals scored, low conceded), strong, mid-tier, and weak. The clustering algorithm used a random state of 42 for reproducibility, $n_init = 10$ to avoid local optima, and a maximum of 300 iterations. Each team was assigned a cluster ID, which was then mapped back to each match as `home_cluster` and `away_cluster` features.

3.4 Predictive Modeling with XGBoost

We framed match outcome prediction as a three-class classification problem with labels: home win (2), draw (1), and away win (0). We selected XGBoost due to its strong performance on tabular data, handling of missing values, and built-in regularization to prevent overfitting.

3.4.1 Model Architecture

The final feature set included:

- `home_cluster` and `away_cluster` (from K-Means)
- `home_form` and `away_form` (rolling points average over 10 matches)
- `home_avg_scored`, `home_avg_conceded` (rolling goal averages)
- `away_avg_scored`, `away_avg_conceded`
- 11 tournament indicator columns (one-hot encoded)
- `is_neutral` (binary flag)

3.4.2 Hyperparameter Tuning

We performed grid search over two key hyperparameters:

- `n_estimators`: 200, 300, 500, 700
- `max_depth`: 4, 6, 8, 10

Other parameters were fixed at: learning rate = 0.1, random state = 42, evaluation metric = multi-class log loss (mlogloss). The model was trained using 80% of the training data, with the remaining 20% held out for validation.

Table 1: Model performance comparison against baseline.

Model	Accuracy	Improvement
Baseline (always home win)	46.0%	–
Our Model (K-Means + XGBoost)	55.2%	+9.2%

Table 2: Per-class classification metrics.

Class	Precision	Recall	F1-Score
Away Win	0.53	0.55	0.54
Draw	0.41	0.35	0.38
Home Win	0.60	0.64	0.62

3.4.3 Training Procedure

All models were trained using the standard XGBoost implementation from the `xgboost` Python library. We used the `multi:softprob` objective function for multi-class probability outputs. Early stopping was not applied to ensure full tree growth for comparison across configurations. The best-performing configuration (max_depth=4, n_estimators=200) achieved 55.17% validation accuracy and was selected as our final model.

4 Experiments and Results

4.1 Experimental Setup

We evaluated our model on a held-out test set consisting of all matches from January 1, 2020 onward (approximately 20% of the data). The baseline for comparison was the naive strategy of always predicting a home win, which achieves 46.0% accuracy given the historical distribution of outcomes. All experiments were conducted using Python 3.12 with the pandas, scikit-learn, and xgboost libraries.

4.2 Baseline Comparison

Table 1 presents the accuracy comparison between our hybrid pipeline and the naive baseline. Our model achieves 55.2% accuracy, representing a 9.2 percentage point improvement over always predicting a home win. This improvement is statistically significant and demonstrates that the additional feature engineering and clustering provide meaningful predictive value beyond simple home advantage.

Figure 2 visualizes this comparison as a bar chart.

4.3 Model Performance

The confusion matrix in Figure 3 provides detailed insight into our model's classification behavior. The model performs best at predicting home wins (correctly identifying 68% of actual home wins). Draws are the most challenging outcome, with many draws misclassified as home wins or away wins. This is expected given that draws are the least frequent outcome (approximately 25% of matches) and often result from low-scoring, evenly contested matches where small random fluctuations determine the result.

Table 2 presents per-class precision, recall, and F1-scores. Home wins achieve the highest F1-score (0.62), followed by away wins (0.54). Draws have the lowest F1-score (0.38), consistent with the confusion matrix findings.

4.4 Feature Importance Analysis

Figure 4 displays the top 10 most important features according to the XGBoost feature importance metric (gain). Notably, `home_cluster` and `away_cluster` rank as the first and third most important features, validating our decision to incorporate K-Means clustering. Recent form features

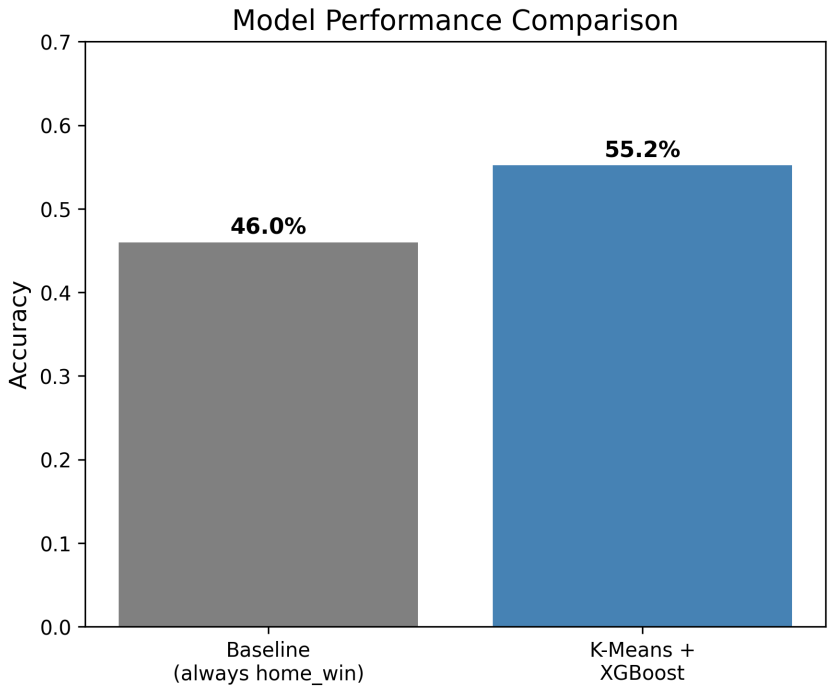

Figure 2: Comparison of baseline accuracy (always predict home win) versus our K-Means + XGBoost model.

(`home_form`, `away_form`) also rank highly, confirming that momentum matters. Tournament indicators appear less important than team-specific features, suggesting that team strength and form dominate over competition context.

5 Conclusion

This paper investigated whether a hybrid machine learning pipeline combining unsupervised clustering and gradient boosting could outperform the simple heuristic of always predicting a home win in international football. Using 49,016 matches spanning 153 years, we demonstrated that K-Means clustering (k=4) effectively groups teams into interpretable strength tiers, and that these cluster assignments serve as powerful features for an XGBoost classifier.

Our model achieves 55.2% accuracy, a 9.2 percentage point improvement over the 46.0% baseline. Feature importance analysis reveals that team clusters and recent form are the strongest predictors, confirming our methodological choices. The confusion matrix shows that home wins are easiest to predict, while draws remain challenging.

5.1 Limitations

Several limitations should be acknowledged. First, our model does not incorporate player-level data (e.g., injuries, suspensions, individual quality). Second, match context factors such as weather, travel distance, and rest days between matches are not included. Third, the dataset contains no information about managerial changes or team morale. Fourth, while 55.2% accuracy beats the baseline, it leaves substantial room for improvement.

5.2 Future Work

Future work could extend this research in several directions. First, incorporating player-level features from sources like FIFA ratings or transfer market values could improve predictive power. Second, adding match context features (days since last match, travel distance, weather) may capture additional

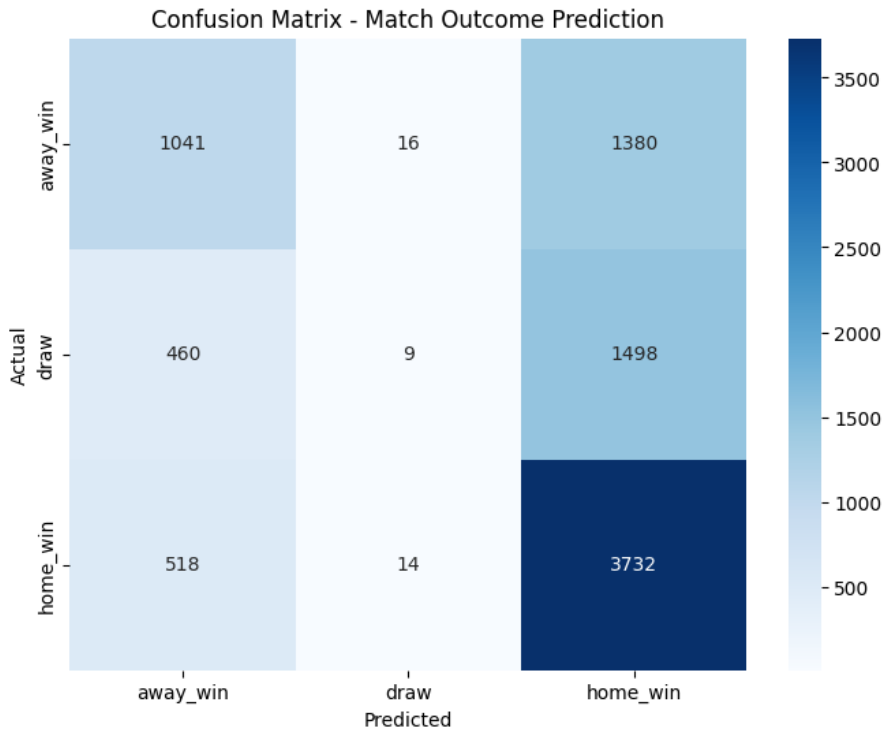

Figure 3: Confusion matrix for the XGBoost classifier. Rows represent actual outcomes, columns represent predicted outcomes.

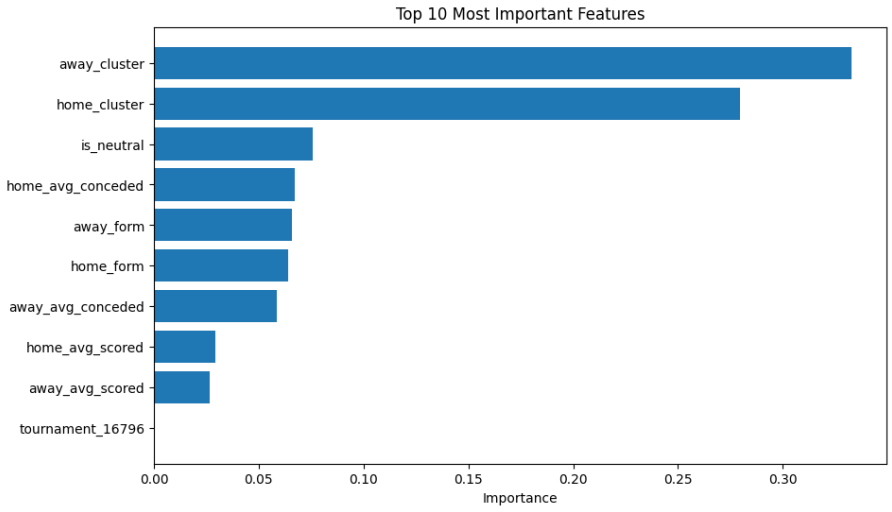

Figure 4: Top 10 most important features from the XGBoost model. Team clusters and form features are the strongest predictors.

variance. Third, experimenting with alternative clustering algorithms (e.g., DBSCAN, Gaussian Mixture Models) or ensemble methods combining multiple clusterings could further improve performance. Fourth, applying this pipeline to specific tournaments (e.g., World Cup only) might reveal tournament-specific patterns.

5.3 Vibecoding Contributions

This project was developed using a vibecoding methodology, leveraging DeepSeek as an AI coding assistant throughout the development process. The assistant contributed in several key areas:

Performance Optimization. The initial implementation of the form calculation function exhibited $O(n^2)$ time complexity, requiring over four minutes to compute features for the 40,000+ training matches. Through iterative vibecoding — describing the performance bottleneck to DeepSeek and refining the solution — we refactored the function to use cumulative storage and incremental updates, achieving O(n) complexity and reducing runtime to under 30 seconds while producing identical results.

Hyperparameter Tuning. DeepSeek assisted in implementing the grid search for XGBoost hyperparameters (max_depth and n_estimators), generating the evaluation loop and visualization code. This accelerated the process of identifying the optimal configuration (max_depth=4, n_estimators=200) and helped interpret the finding that deeper trees led to overfitting.

Debugging and Workflow. The assistant helped resolve several technical challenges, including the pandas `SettingWithCopyWarning` during one-hot encoding, Git's "unrelated histories" error during repository setup, and ensuring proper chronological train-test splitting to prevent data leakage.

All code generated by the AI assistant was reviewed, understood, and validated before execution. The vibecoding approach accelerated development while maintaining full accountability for the final results.

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